

Q.) Trace the following operations in Circular Double Linked List.

(i) Insert 46, 93, 76, 35 in the beginning.

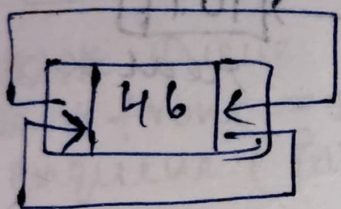
(ii) Insert after 76.

Insert at the end 100

Insert at the end 200
300
400

(iii) Delete 100, 400, 76.

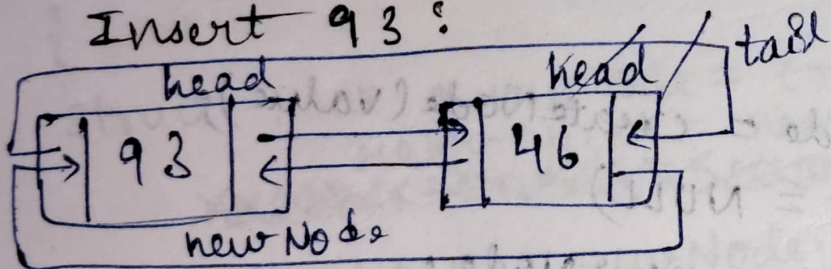
i) Insert 46 at beginning?



head/tail

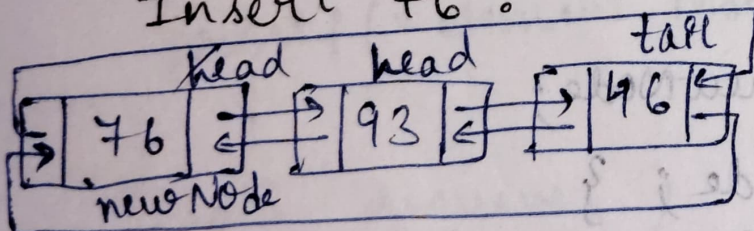
```
if (head == NULL) {
    head = newNode;
    tail = newNode;
    head->next = head;
    head->prev = head;
}
```

Insert 93?

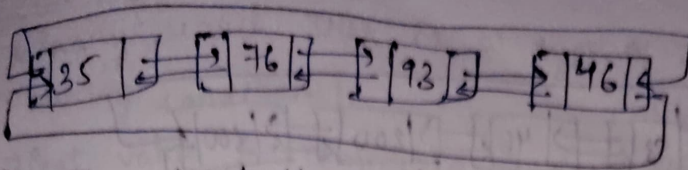


```
else {
    newNode->next = head;
    newNode->prev = tail;
    head->prev = newNode;
    tail->next = newNode;
    head = newNode;
}
```

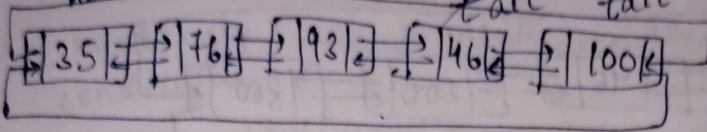
Insert 76?



Insert

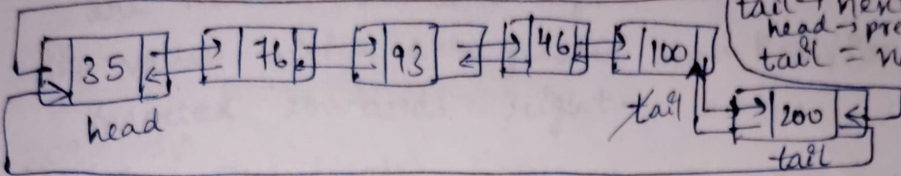


(ii) Insert at the end 100 %

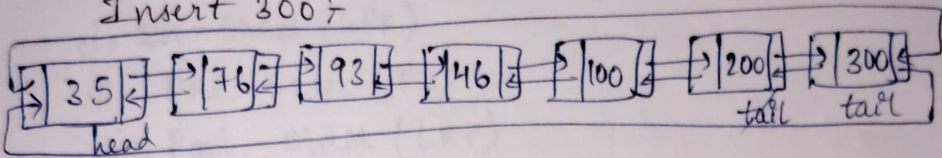


newNode $\rightarrow$ prev = tail;
newNode $\rightarrow$ next = head;
tail $\rightarrow$ next = newNode;
head $\rightarrow$ prev = newNode;
tail = newNode;

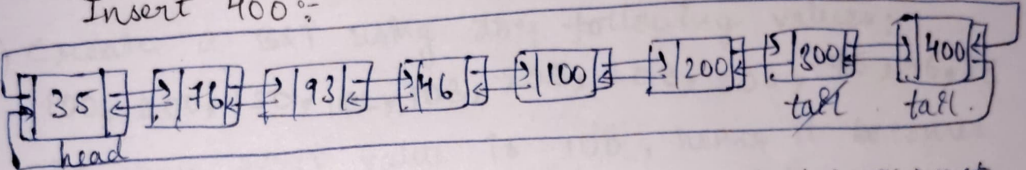
Insert 200 :



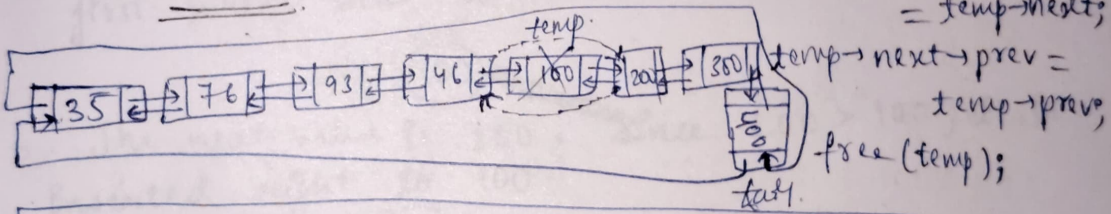
Insert 300 :



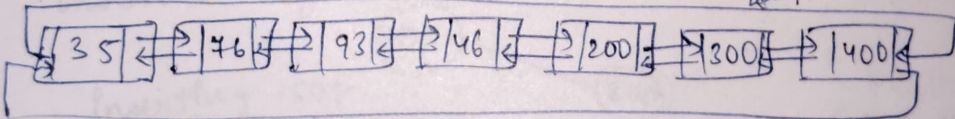
Insert 400 :-



Delete 100 :-



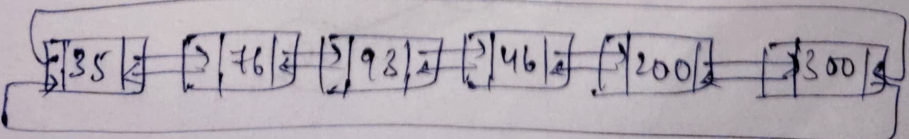
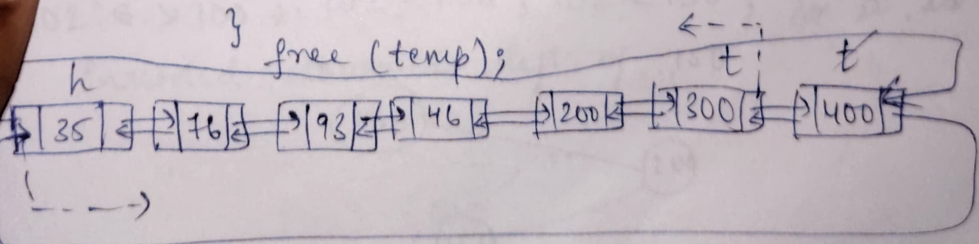
temp $\rightarrow$ prev $\rightarrow$ next
= temp $\rightarrow$ next;
temp $\rightarrow$ next $\rightarrow$ prev =
temp $\rightarrow$ prev;
free (temp);



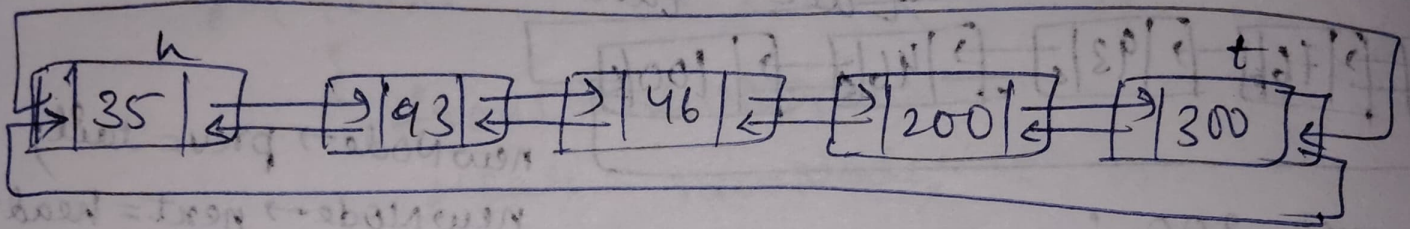
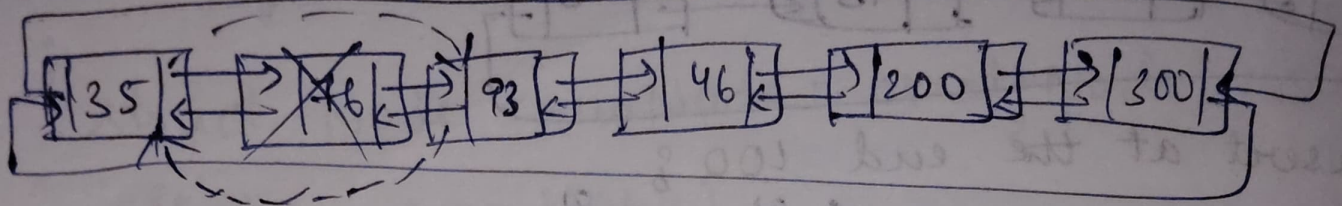
Delete 400 : Delete at end.

tail = tail $\rightarrow$ prev;
tail $\rightarrow$ next = head;
head $\rightarrow$ prev = tail;

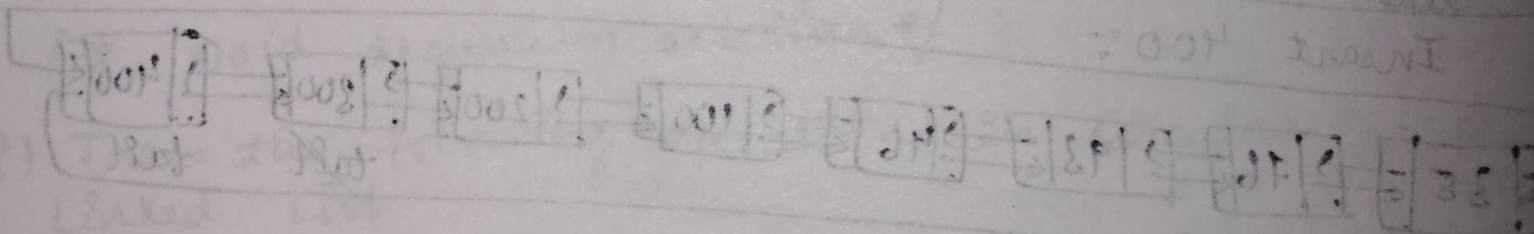
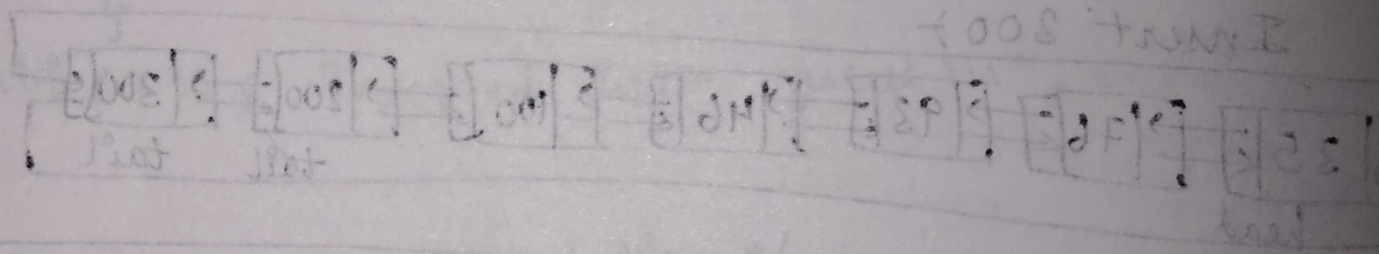
} free (temp);



Delete 76 :-



temp = temp->next
temp->next = new node
temp->next->next = temp->next->next



temp = temp->next
temp->next = temp->next->next
temp->next->next = temp->next->next

